SECTION 1

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. FLABBY: (A) fertile (B) taut (C) radiant (D) fortunate (E) miniature

2. CONCLUSIVE: (A) debatable (B) tactless (C) obvious (D) aggravating (E) noisy

3. INERT: (A) exact (B) dry (C) active (D) congenital (E) hopeful

4. HEARTEN: (A) assume (B) dismay (C) imitate (D) confide in (E) sympathize with

5. OBSCURITY: (A) prominence (B) continuity (C) reliability (D) gentleness (E) resistance

6. ADULTERATE: (A) unite (B) conclude (C) polish (D) purify (E) impute

7. ZENITH: (A) slow progress (B) hidden side (C) distant object (D) revealed truth (E) lowest point

8. CHERUBIC: (A) ecstatic (B) provincial (C) very colorful (D) highly potent (E) fiendish

9. ARTFUL: (A) forlorn (B) attentive (C) guileless (D) oppressive (E) stubborn

10. GRIEVOUS: (A) minor (B) visible (C) incomplete (D) inevitable (E) genuine

11. AMITY: (A) animosity (B) tragedy (C) poverty (D) sobriety (E) ordeal

12. AMELIORATE: (A) aggravate (B) forget (C) refuse (D) hesitate (E) amputate

13. EPHEMERAL: (A) valiant (B) legible (C) apathetic (D) predestined (E) eternal

14. BROACH: (A) devalue (B) close (C) reprove (D) affirm (E) heed

15. IRASCIBLE: (A) honorable (B) fickle (C) delectable (D) stubborn (E) impassive

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Because mist is a magician, even the ---- object may appear beautiful under certain atmospheric conditions.

(A) largest (B) heaviest (C) ugliest (D) simplest (E) strongest

17. The great voyages of discovery that expanded people's geographic vision served to ---- their intellectual horizons as well.

(A) engulf (B) immobilize (C) clarify (D) broaden (E) fragment

18. Rebecca's ---- of ---- in all its forms was consistent with her commitment to reason and moderation.

(A) dismissal. .doubt
(B) suspicion. .fanaticism
(C) disparagement. .frailty
(D) tolerance. .intemperance
(E) exaggeration. .courtesy

19. Some critics maintain that Mahler's music is uneven, running the gamut from the --- to the great.

(A) unprecedented (B) monumental (C) essential (D) sublime (E) banal

20. The foreign correspondent portrayed the leaders of both warring factions so sympathetically that he was ---- by both sides for his blindness to the enemy's ----.

(A) admonished. .virtue
(B) censured. .affability
(C) commended. .integrity
(D) vilified. .baseness
(E) revered. .innocence

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Can poverty in the United States be abolished within the limits of the welfare state?

The answer is clear enough. The government's own figures demonstrate that the current antipoverty programs are basically inadequate. I do not, however, want to dismiss completely the government's anti-poverty programs. Current serious discussion of poverty in this country is a gain which one owes in part to that program. But there is no point in pretending that a little more welfarism will do away with a national shame.

Today's poor are different from the pre-Second World War poor. The "old" poor lived at a time when economic opportunity was the national trend, when the net income from the growth of American manufacturing increased by 4,500 per cent. It was the "old" poor, mostly Eastern European immigrants unified by language and culture, who created the big-city political machines and participated in the organization of unions and the political struggle for the New Deal. They had objective, realistic reason for hope.

An analysis of the first phase and second phase of the New Deal of the Roosevelt administration is quite relevant at this point. The first phase of the New Deal, supported by American business and dominated by the National Recovery Act, gave recognition to an old corporate dream—economy planned by business. The second phase of the New Deal (the source of today's welfare theory and antipoverty wisdom) moved away from the concept of planning and toward a "free market." The assumption was that in its intervention the government should not plan but should stimulate the economy and that the private sector and initiative would continue to be the mainspring of progress.

After the Second World War, the government started emphasizing training programs because some workers were not participating in the general economic advance. However, these training programs have missed the fundamental problem. The novelty of impoverishment today is that it takes place in a time of automation. The government offers education and training and at the same time admits that the jobs for its graduates are obsolete. Such hypocrisy reinforces the cynicism and resistance to organization which characterizes poor communities.

It is therefore crucial that the federal government generate jobs and create an environment of economic hope. The essence of the "third phase" of the New Deal would be social investment, a conscious and political allocation of resources to meet public needs. This New Deal would be dependent upon a coalition, which would include, but not be confined to, the poor, that would see to it that planning and social investment were extended in a democratic way.

21. It can be inferred that the first phase of the New Deal

(A) discouraged investment in industry
(B) was the source of today's welfare theory
(C) stimulated the growth of big-city political machines
(D) provided programs to prepare workers for jobs in automated industry
(E) gave business an opportunity to extend control over the economy

22. The author credits the government's antipoverty program with which of the following?

(A) Encouraging serious discussions on national poverty
(B) Promoting a new welfare system
(C) Establishing programs to generate jobs
(D) Eliminating a national shame
(E) Involving the poor in plans for social reform

23. The primary purpose of the passage is to

(A) expose those who support government anti-poverty programs
(B) distinguish between the pre- and post-Second World War poor
(C) argue for social investment by the federal government to alleviate poverty
(D) reveal the practices of society that perpetuate poverty
(E) distinguish among the first, second, and third New Deal approaches to reducing unemployment

24. According to the passage, today's poor differ from the "old" poor in which of the following ways?

I. The "old" poor enjoyed a higher average income.
II. The "old" poor had a realistic hope for economic opportunity.
III. The "old" poor were better educated.

(A) II only (B) III only (C) I and III only
(D) II and III only (E) I, II, and III

25. The author's attitude toward the government's current poverty program can best be described as

(A) unqualified appreciation
(B) fundamental dissatisfaction
(C) apathetic acceptance
(D) self-righteous pomposity
(E) violent indignation

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

Scientists studying choriocarcinoma, a kind of cancer developing out of fetal tissue, are inevitably students of the fetus-mother relationship and of the phenomenon by which a mother's body suppresses normal rejection of foreign tissue such as the fetus itself. Many scientists feel that the trophoblast is responsible for this phenomenon. The trophoblast that gives rise to choriocarcinoma first appears as a layer of cells covering the blastocyst (an early stage of the embryo) that is located on the surface of the placenta. It penetrates and erodes the surface tissue of the uterine wall with the result that the blastocyst becomes attached to the uterus. Moreover, the eroding action of the trophoblast damages and thus opens up some of the uterine blood vessels, providing a source of nourishment for the fetus. It appears obvious that the trophoblast has some peculiar property which permits its toleration by the mother's body. Especially important may be the noncellular substance called fibrinoid (or the layer of Nitabuch) covering the cells of the trophoblast. It has recently been hypothesized that the layer of Nitabuch, known for at least a hundred years, may be a barrier to prevent the escape of antigens from the embryonic tissues so that the mother's immune response cannot be stimulated into taking action against the foreign tissue growing in her uterus.

There is a lot of evidence to suggest that in the normal rejection process some sort of direct cell-to-cell contact must occur, and that the host cell making direct contact with the foreign tissue is a white blood cell—the lymphocyte. What the lymphocyte does when it establishes contact is not clear. Some message goes back to the lymph gland in the area, with the result that there is a proliferation of lymphocytes in the lymph gland. These are specialized lymphocytes that can then move in to attack the foreign tissue; and there may also be a secondary process in which antibodies are produced. But if the lymphocyte (which has a fairly high net negative surface charge) opposed a cell that also had a high negative surface charge, then it might not be possible for the lymphocyte to make the appropriate contact, since negative charges repel. It is difficult to measure the electrical charge of the trophoblast cell, but it is likely that it has a high net negative charge, perhaps due to the layer of Nitabuch. If the lymphocyte is prevented from making contact with the cell of the foreign body, then the foreign body—be it fetus or the fetus-originated choriocarcinoma—will not be rejected.

26. Which of the following can be inferred about choriocarcinoma?

(A) Only embryos are immune to its attack.
(B) It is most commonly associated with pregnancy.
(C) It is the type of cancer which has been most extensively studied.
(D) It is the only type of cancer for which the body's rejection process is important.
(E) It is the only type of cancer for which a cure has been found.

27. The author states that foreign tissue will not be rejected by the host if

(A) the layer of Nitabuch has been damaged
(B) the cells of the host are positively charged
(C) the host is already fighting off other foreign cells
(D) direct contact between a lymphocyte and a foreign cell is prevented
(E) the structure of the foreign cells resembles that of the host cells

28. The author provides information to answer which of the following questions?

(A) Where do the body's lymphocytes originate?
(B) What is the process by which antibodies are produced?
(C) What does a lymphocyte do after making contact with a foreign cell?
(D) What is the mechanism by which the layer of Nitabuch prevents the escape of antigens from embryonic tissue?
(E) What are the issues on which scientists disagree concerning the role of the trophoblast?

29. The author's discussion suggests that

(A) cancer research has accomplished very little
(B) much of the explanation he offers is only tentative
(C) current cancer research emphasizes prevention rather than cure
(D) he is not qualified to judge the significance of the research he describes
(E) many scientists have overemphasized study of the fetus-mother relationship to the exclusion of other aspects of cancer research

30. Which of the following titles best describes the content of the passage?

(A) The Layer of Nitabuch—Cause of Choriocarcinoma
(B) The Lymphocyte and Its Ways of Defeating Disease
(C) The Mutually Beneficial Relationship of the Trophoblast and the Blastocyst
(D) Some Discoveries Revealing Possible Cures for Choriocarcinoma
(E) Some Speculations as to Why the Mother's Body Fails to Reject Choriocarcinoma

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. Many great dramatic parts make such heavy --- the physical and spiritual resources of the chief performers that eight performances a week are an ---- strain.

(A) impositions on. .appropriate
(B) requirements on. .invigorating
(C) overtures to. .ultimate
(D) demands on. .unbearable
(E) competition for. .unmitigated

32. In their search for artifacts, archaeologists are often ----, not because a suspected site is remote and isolated but because it is ----.

(A) misled. .verified
(B) undeterred. .unearthed
(C) belittled. .undespoiled
(D) venerated. .sacred
(E) frustrated. .urbanized

33. The ---- of the individual's desire for freedom despite constant efforts to destroy it does not mean that freedom will triumph; at best, it suggests that the desire for it is ----.

(A) repudiation. .enviable
(B) vacillation. .insatiable
(C) emasculation. .universal
(D) reassertion. .unforeseen
(E) persistence. .indestructible

34. The ---- of erecting arbitrary barriers between the sciences is clearly demonstrated in the study of sensory processes, where progress has been made through the ---- efforts of many specialists.

(A) efficacy. .qualified
(B) utility. .united
(C) possibility. .diverse
(D) irrationality. .combined
(E) practice. .immense

35. Naturally his course brought down on him the --- of both leftists and moderates, and he had long since ---- the conservative elements of the population.

(A) admiration. .reactivated
(B) anger. .appeased
(C) malediction. .misdirected
(D) respect. .alienated
(E) distrust. .estranged

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. SEED : SPROUT :: (A) lava : volcano (B) bud : bloom (C) fish : minnow (D) pollen : bee (E) vine : tree

37. RUMOR : FACT :: (A) myth : history (B) guess : assumption (C) science : knowledge (D) fiction : story (E) inference : intuition

38. SNARL : ANGRY :: (A) bite : pensive (B) growl : humble (C) purr : contented (D) search : defensive (E) pounce : secretive

39. DERIVATION : WORD :: (A) ancestry : person (B) nomenclature : term (C) address : family (D) validation : document (E) relationship : brother

40. LONESOME : COMPANIONSHIP :: (A) nervous : agitation (B) bored : stimulation (C) clumsy : determination (D) suspicious : caution (E) arrogant : ambition

41. NEEDLE : SEW :: (A) padlock : enter (B) picture : draw (C) gauge : measure (D) sound : fear (E) rope : knot

42. EMENDATION : TEXT :: (A) introduction : speech (B) boundary : territory (C) alteration : garment (D) sketch : painting (E) dilapidation : building

43. HEMORRHAGE : BLOOD :: (A) battle : wound (B) deterrence : aggression (C) consensus : opinion (D) opulence : wealth (E) effusion : emotion

44. UNREMITTING : CESSATION :: (A) unique : corruption (B) irrelevant : distraction (C) uniform : diversity (D) unbecoming : permanence (E) indisputable : certainty

45. DISFRANCHISEMENT : RIGHTS :: (A) corruption : ability (B) emancipation : freedom (C) imitation : originality (D) organization : individuality (E) enervation : strength

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

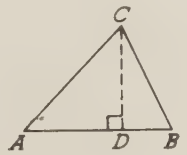
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

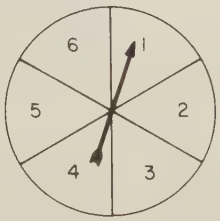
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1. The cost of 900 items at \$1.65 per 100 is

   (A) \$14.85
   (B) \$18.30
   (C) \$148.50
   (D) \$183.00
   (E) \$1,485.00

2. $51(52 + 53 + 54)$ is equal to all of the following EXCEPT

   (A) $51(52) + 51(53 + 54)$
   (B) $51(52) + 51(53) + 51(54)$
   (C) $(54 + 53 + 52)51$
   (D) $(54 + 53)51 + 52(51)$
   (E) $51 + (52)(53)(54)$



3. In a certain game, a player gets an additional spin if, when he spins the arrow on the disc shown above, the arrowhead stops in the region numbered 6. On a given spin, what is the chance that he will get an additional spin? (All regions are of equal size.)

   (A) 1 in 36
   (B) 1 in 6
   (C) 1 in 5
   (D) 1 in 3
   (E) 1 in 2

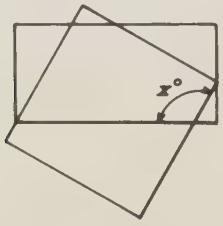
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4. If $y - 4 = 9$, then $y + 4 =$

(A) 4 (B) 8 (C) 9 (D) 13 (E) 17

5. If $x > 0$ and $x^2 - 1 = 15 \times 17$, then $x =$

(A) 11
(B) 14
(C) 15
(D) 16
(E) 17



6. In the figure above, a square and a rectangle overlap. Which of the following is equal to x ?

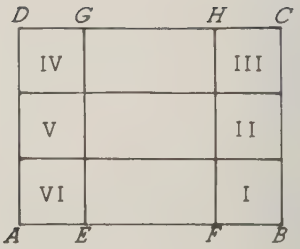
(A) 120 (B) 135 (C) 140 (D) 150
(E) It cannot be determined from the information given.

7. A load of 1,000 tons on a truck is increased by one per cent. What is the weight, in tons, of the adjusted load?

(A) 1,001
(B) 1,010
(C) 1,100
(D) 1,110
(E) 1,111

8. If $6 + x + y = 20$ and if $x + y = k$, then $20 - k$ is equal to

(A) 14 (B) 9 (C) 6 (D) $3\frac{2}{3}$
(E) none of the above



9. $ABCD$ is a rectangle and squares I through VI have equal areas. If $EF = 2AE$, the area of one of the squares is what fraction of the area of rectangle $ABCD$?

(A) $\frac{1}{12}$ (B) $\frac{1}{6}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{2}{3}$

10. Which of the following fractions is the greatest?

(A) $\frac{16}{17}$
(B) $\frac{8}{9}$
(C) $\frac{14}{15}$
(D) $\frac{9}{10}$
(E) $\frac{11}{12}$

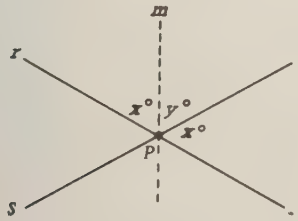
11. A woman made 5 payments on a loan with each payment being twice the amount of the preceding one. If the total of all 5 payments was \$465, how much was the first payment?

(A) \$5
(B) \$15
(C) \$31
(D) \$93
(E) \$155

GO ON TO THE NEXT PAGE

12. If $20 \cdot 3{,}000 = 6 \cdot 100^x$, then $x =$

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6



13. In the figure above, line r intersects line s at P. If another line m is drawn through P with angles formed as indicated in the figure, then what is the value of y in terms of x ?

(A) 60 (B) x (C) $2x$
(D) $180 - x$ (E) $180 - 2x$

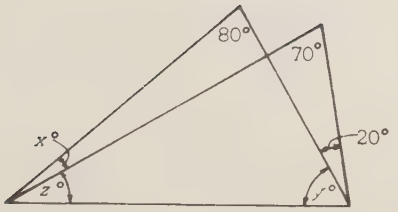
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14. If $k = \frac{n}{12} + \frac{n}{12} + \frac{n}{12}$, then the least positive integer n for which k is an integer is

(A) 2
(B) 3
(C) 4
(D) 6
(E) 8

15. How many integers are both greater than the integer n and less than $n + 1$?

(A) None
(B) One
(C) n
(D) $n + 1$
(E) It cannot be determined from the information given.



16. In the figure above, if $z = 3x$, then $y =$

(A) 30 (B) 60 (C) 70 (D) 80 (E) 90

---

17. If N is the least positive integer for which $3N$ is both an even integer and equal to the square of an integer, then $N =$

(A) 3 (B) 4 (C) 6 (D) 12 (E) 18

18. Bud now weighs 9 kilograms more than he did a year ago. If his weight then, in kilograms, was $\frac{9}{10}$ of his present weight, what is his present weight?

(A) 72 kg
(B) 81 kg
(C) 90 kg
(D) 99 kg
(E) 108 kg

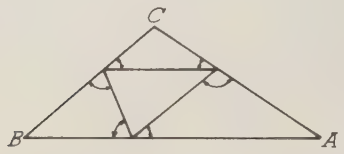
19. The volume of a cube with edge $\sqrt{2}$ is how many times the volume of a cube with edge $\sqrt[3]{2}$?

(A) $\sqrt[3]{2}$
(B) $\sqrt{2}$
(C) 2
(D) $2\sqrt{2}$
(E) 4

GO ON TO THE NEXT PAGE

20. For any integer n, which of the following represents three consecutive even integers?

(A) $n, n+1, n+2$
(B) $n, n+2, n+4$
(C) $2n, 4n, 6n$
(D) $2n, 2n+1, 2n+2$
(E) $2n, 2n+2, 2n+4$



21. In the figure above, if ABC is a triangle, what is the sum of the degree measures of the marked angles?

(A) 180°
(B) 270°
(C) 360°
(D) 450°
(E) It cannot be determined from the information given.

22. The height of triangle T is 50 per cent of the width of rectangle R, and the length of the base of triangle T is 40 per cent of the length of rectangle R. If the area of triangle T is 3, what is the area of rectangle R ?

(A) 1
(B) 10
(C) 15
(D) 20
(E) 30

23. If 3 blots equal 4 bleets and 5 bleets equal 6 blits, what is the ratio of one blit to one blot?

(A) $\frac{1}{2}$
(B) $\frac{5}{8}$
(C) $\frac{2}{3}$
(D) $\frac{8}{5}$
(E) $\frac{2}{1}$

24. The average of 10 students' test scores is 72. When the 2 highest and 2 lowest scores are eliminated, the average of the remaining scores is 68. What is the average (arithmetic mean) of those eliminated?

(A) 64
(B) 76
(C) 78
(D) 80
(E) It cannot be determined from the information given.

25. If N is a positive whole number, let N^* equal the sum of all whole numbers from 1 to N, inclusive; for example, $6^* = 6+5+4+3+2+1 = 21$. Which of the following are true?

I. 20^* is an odd number.
II. If P is an odd whole number, then P^* is odd.
III. $(R+1)^* - R^*$ is equal to $R+1$.

(A) None
(B) I only
(C) II only
(D) III only
(E) II and III

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 4

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. PERISHABLE:
   (A) immobile
   (B) inexcusable
   (C) unable to advance
   (D) unlikely to deteriorate
   (E) unaccustomed to routine

2. DRAWBACK: (A) similarity (B) advantage (C) prediction (D) evidence (E) introduction

3. OUTPOST: (A) central location (B) large crowd (C) strong defense (D) vital fact (E) active leadership

4. PULVERIZE: (A) mold together (B) gain weight (C) make narrow (D) use force (E) push aside

5. DAWDLE: (A) liberate (B) soothe (C) brighten (D) solidify (E) rush

6. GRAPHIC: (A) politely refused (B) gracefully executed (C) poorly delineated (D) unintentional (E) modest

7. SQUANDER: (A) idealize (B) economize (C) classify (D) move rapidly (E) agree meekly

8. REVERE: (A) despise (B) restrain (C) misunderstand (D) conceal deliberately (E) leave unchanged

9. SACCHARINE: (A) taciturn (B) elated (C) vitriolic (D) amicable (E) squalid

10. RECALCITRANT: (A) obedient (B) energetic (C) temporary (D) perceptive (E) conservative

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. When, in 1972, the Indian government found that there were only 1,827 tigers left, the shooting of tigers was ----, and steps were taken to preserve their ----.
   (A) prohibited. .habitats
   (B) limited. .examination
   (C) investigated. .multiplication
   (D) reduced. .nature
   (E) legalized. .well-being

12. Ms. Cartright's arguments are so well organized, clearly expressed, and extensively documented that even the most zealous of her opponents have ---- the --- of her stance.
   (A) recognized. .fallacy
   (B) minimized. .legitimacy
   (C) garbled. .frankness
   (D) acknowledged. .merit
   (E) modified. .ambiguity

13. Neither Green nor Simmons had particularly --- ambitions; both seem to have been concerned more with the --- of their audience than with the pursuit of truth.
   (A) humanitarian. .virtues
   (B) laudable. .plaudits
   (C) philanthropic. .needs
   (D) ignoble. .blandishments
   (E) materialistic. .treasures

GO ON TO THE NEXT PAGE

14. There is an effort afoot among dissatisfied scientists to ---- what has clearly been an imbalance in the fields chosen for intensive research.

(A) augment (B) endorse (C) underbid
(D) reinstate (E) redress

15. His characteristic dogmatism is occasionally -- by his ability to see merit in the ideas of others.

(A) rationalized (B) intensified
(C) substantiated (D) disqualified
(E) redeemed

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. COMPOSER : PERFORMER :: (A) painter : wall
(B) poet : posterity (C) intellectual : lowbrow
(D) playwright : actor (E) philosopher : eternity

17. CAPILLARY : ARTERY :: (A) bridge : gorge
(B) bank : river (C) ebb : flow
(D) twig : branch (E) blood : heart

18. TASKMASTER : CREW :: (A) director : cast
(B) soloist : chorus (C) observer : crowd
(D) prisoner : jury (E) truant : class

19. PAINSTAKING : CAREFUL ::
(A) restive : destructive (B) generous : polite
(C) punctual : busy (D) engrossed : disturbed
(E) fastidious : neat

20. DAM : WATER :: (A) ramp : turnpike
(B) tunnel : mountain (C) dock : ships
(D) roadblock : traffic (E) balloon : air

21. ADJOINING : LOCATION ::
(A) consecutive : time (B) coincidental : error
(C) occasional : duration (D) accidental : purpose
(E) vacant : space

22. LARCENY : PROPERTY :: (A) arson : building
(B) libel : slander (C) plagiarism : ideas
(D) counterfeit : evidence (E) forgery : pen

23. LABYRINTH : CONFUSION ::
(A) deadlock : arbitration (B) catalyst : delay
(C) treadmill : boredom (D) pattern : chaos
(E) riddle : solution

24. PARENTHESIS : PARAGRAPH ::
(A) illness : sanity (B) conclusion : experiment
(C) stage : dialogue (D) knowledge : wisdom
(E) digression : argument

25. PARIAH : SOCIETY :: (A) novice : sport
(B) exile : country (C) leper : colony
(D) tyrant : dictatorship (E) chairman : committee

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

We were four companions. Unforeseen chance had joined us together on a large sugar plantation on the Peruvian coast. We worked at different occupations during the day and met during the evening. Not being Englishmen, we did not play cards. Instead our constant discussions led to disputes. These didn't stop us from wanting to see each other the next night, however, to continue the interrupted debates and support them with new arguments. Nor did the rough sentences of the preceding wrangles indicate a lessening of affection, of which we assured ourselves reciprocally with the clasping of hands. On Sundays we used to go on hunting parties. We roamed the fertile glens, stalking, generally with poor results, the game of the warm region around the coast.

We came to be tireless wanderers and excellent marksmen. What attracted us was the trans-Andean region: fertile plateaus extending on the other side of the range in the direction of the Atlantic toward the immense land of Brazil. It was as if primitive nature called us to her breast. The vigor of the fertile, untouched jungles promised to rejuvenate our minds, the same vigor which rejuvenates the strength and thickness of the trees each year. At times we devised crazy plans. As with all things that are given a lot of thought, these schemes generally materialized. Ultimately nature and events are largely what our imaginations make them out to be. At the end of the year, with arranged vacations, accumulated money, good rifles, and four hammocks, our caravan descended the Andean slopes leading to the endless green ocean.

26. The four companions in the passage are discussed primarily in terms of

(A) the topics they discussed
(B) their mutual activities
(C) their interaction with others
(D) the characteristics of the land in which they lived
(E) the different personalities of each

27. It can be inferred from the passage that the acquaintanceship among the four companions developed because they

(A) held similar views on most issues
(B) were all of the same nationality
(C) worked in the same general location
(D) had trained in the same profession
(E) had traveled together previously

28. The companions seemed to view the jungle as a source of

(A) mysterious danger
(B) impassive peace
(C) lucrative profits
(D) inconsequential diversion
(E) refreshing vitality

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4

The assumption that chlorofluorocarbons would be innocuous in the environment because they were chemically inert might have gone unchallenged for some time, but at a conference someone mentioned their curious persistence in the atmosphere to F. Sherwood Rowland, a chemist, who began to wonder what ultimately became of the synthetic compounds. In 1973, Rowland got a research grant to study the question, and he and his associate, Mario J. Molina, set out to prove that chlorofluorocarbons must be floating unchanged to the stratosphere, where the strong ultraviolet rays in sunlight would cause them to release chlorine atoms. These atoms would act as catalysts to destroy ozone, with each chlorine atom capable of setting off a chain reaction involving thousands of ozone molecules. These ozone molecules are part of the atmospheric ozone layer that protects the earth from solar ultraviolet rays. It is thought that if this layer is depleted, the results could be increased incidences of skin cancer, damage to plant life, and even global climatic changes.

Even though the depletion of the ozone layer would be lessened if production of the compounds ceased immediately, Rowland and Molina concluded that a twenty to forty per cent reduction of the layer was a strong possibility. The most disturbing conclusion was that the full effect of chlorofluorocarbons already in the atmosphere would not be felt for at least a decade, since the gases floated up and down until they gradually ascended to the ozone layer. "There was no moment of triumph," Rowland recalled. "I just came home one night and told my wife, 'The work is going very well, but it looks like the end of the world.'"

29. The passage primarily concerns the

(A) effects of ozone on plant and animal life
(B) interaction of ultraviolet rays with ozone
(C) potentially dangerous uses for chlorofluorocarbons
(D) discovery of the effect of chlorofluorocarbons on the ozone layer
(E) variety of gases in the atmosphere surrounding the earth

30. It can be inferred that before the publication of the study of Rowland and Molina, it was widely assumed that chlorofluorocarbons in the environment

(A) were harmless
(B) absorbed ultraviolet rays
(C) reacted with ozone
(D) were rapidly increasing
(E) could not be reproduced in a laboratory

31. According to the hypothesis of Rowland and Molina, the chlorofluorocarbons would release chlorine atoms when

(A) the ozone broke up the chlorofluorocarbon molecules
(B) they were subjected to the lowered air pressure of the stratosphere
(C) they were exposed to strong ultraviolet rays in the stratosphere
(D) they floated down from the stratosphere closer to the earth's surface
(E) the amount of ozone in the stratosphere was more than twenty per cent

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Present-day Chicanos are the fulfillment of a cosmic cycle from ancient Aztlán (what is now the southwestern part of the United States), the seed ground of the great civilizations of Anahuac, to modern Aztlán. We have rediscovered Aztlán in ourselves. This knowledge provides the dynamic principle on which to build a deep unity and brotherhood among Chicanos. Ties much more profound than even language, birthplace, or culture bind us together—Aztlán represents the unifying force of our nonmaterial heritage.

References to Aztlán as the place of origin of the Mexican Indian peoples are negligible in North American chronicles. Two of the most easily obtainable texts by historians of the United States are Prescott's *History of the Conquest of Mexico* (1843) and Josephy's *The Indian Heritage of America* (1968). Prescott, in reviewing the various histories compiled for the most part by priest-scholars, suggested that the American peoples were descended from one of the lost tribes of Israel. He wrote:

> The theory of an Asiatic origin for Aztec civilizations derives strong confirmation from tradition. Traditions concerning their origin were found among the tribes, and were preserved both orally and in hieroglyphical maps, where the different stages of their migration are carefully noted. But who at this day shall read them? They are admitted to agree, however, in representing the populous North as the prolific hive of the American races. In this quarter were placed their Aztlán and their Huehuetlapallan, the abodes of their ancestors.

In a footnote, he says of the maps, "But as they are all within the boundaries of New Spain, indeed, south of the Rio Gila, they throw little light, of course, on the vexed question of the primitive abodes of the Aztecs."

It so happens that the Rio Gila is part of a convergence of rivers and cultures as significant as the confluence of the Tigris and Euphrates. Yet Prescott would have us seek a more distant source.

In a comprehensive study of the Indians of the Americas, Josephy recounts the settling of the Mexica tribe on the site of today's Mexico City. He says that the Mexica took the name Aztec from Aztlán, whence they had come, "somewhere vaguely to the Northwest and perhaps even in the present-day United States Southwest."

Mexican scholars clearly identify Aztec and Mexican origins with the southwestern United States. However, aside from the two sources cited, further reference to Aztlán is difficult to find in United States histories. It has obviously been of little consequence to anyone except the Chicano. We still know very little about our ancient origins.

32. According to the author, Aztlán today can be described as which of the following?

I. A unifying force
II. An abstract principle
III. An actual location

(A) I only (B) III only (C) I and II only
(D) II and III only (E) I, II, and III

33. According to the passage, Prescott's information about Aztlán was derived primarily from his own

(A) interviews with Mexican Indians
(B) careful reading of Aztec hieroglyphics
(C) study of the writings of earlier historians
(D) field work done in the American Southwest
(E) studies of Indian life-styles

34. The author mentions the works of Prescott and Josephy chiefly because

(A) they are among the few sources that mention Aztlán
(B) their authors are particularly knowledgeable about North American Indians
(C) they stress the significance of Aztlán for the Chicano
(D) they are the texts commonly used in North American schools
(E) they provide the most comprehensive treatments of North American history

35. The tone of Josephy's statement about the location of Aztlán (lines 49-51) can best be described as one of

(A) apologetic regret
(B) disguised irony
(C) cautious speculation
(D) dramatic revelation
(E) philosophical resignation

36. The information in the passage suggests that the author is most likely

(A) a historian who is concerned about the validity of his sources
(B) a Chicano who is interested in bringing the Chicanos together
(C) a literary critic who questions the conclusions of historians
(D) an educator primarily concerned with the future of Chicano children
(E) a researcher who is interested in discovering new facts about the Mexican Indians

GO ON TO THE NEXT PAGE

4

If we are to understand the impact of Aristotle on subsequent thought, we must remember some curious facts connected with the transmission of his writings. When Aristotle died in 322 B.C., he left a very extensive body of writings that consisted of two completely different groups. The first group of writings included a large number of dialogues and other popular treatises, which had been published during his lifetime and which continued to be widely read through many centuries until they were finally lost toward the end of antiquity. These popular writings of Aristotle were praised for their literary elegance, and apparently the most famous among them were composed in Aristotle's earlier years and were comparatively close to Plato in their philosophical opinions. The second group of Aristotle's writings, which is the one that has come down to us, represents a collection of lecture courses that he delivered in his school in Athens. These courses served no immediate literary purpose, but instead are highly technical in character, very detailed in their reasoning and in the information supplied, and fairly systematic in their overall arrangement, forming a vast encyclopedia of philosophical and scientific knowledge. These lectures of Aristotle were not published by him or his immediate successors; they remained for several centuries in the library of his school, where they were accessible to its professors but not to the general public or to the members of other schools of philosophy.

37. Which of the following best summarizes the "curious facts" (lines 2-3) that the author focuses on in the passage?

(A) Few of Aristotle's contemporaries were and no modern scholars are familiar with both groups of his writings.
(B) Aristotle's academic lectures were no less elegantly written than were his more famous dialogues and treatises.
(C) The content of Aristotle's popular dialogues is contradicted in the lecture courses he delivered in his school.
(D) The popularity of Aristotle's writings ceased when he died and it was not revived for many centuries.
(E) The content of Aristotle's academic lectures was popularized not by his followers but by members of other Athenian schools of philosophy.

38. It can be inferred that by "transmission" (line 3) the author means

(A) inspiration or motivation for action
(B) analysis or interpretation of theory
(C) alteration or modification of material
(D) communication or dissemination of information
(E) relocation or replacement of evidence

39. The author appears most certain of the answer to which of the following questions?

(A) Why were copies of Aristotle's popular treatises lost?
(B) How is the influence of Plato evident in Aristotle's earlier treatises?
(C) What was the content of the lecture courses given by Aristotle at his school?
(D) Why did Aristotle not publish the lectures he gave at his school?
(E) What was the content of Aristotle's popular treatises?

40. The author's attitude toward Aristotle's writings is best described as one of

(A) unqualified endorsement
(B) apologetic approval
(C) analytical objectivity
(D) skeptical reserve
(E) scholarly dissatisfaction

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

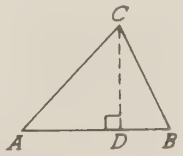
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $x = 8$ and $y = 6$, then $\dfrac{\frac{x}{2} + \frac{y}{3}}{\frac{xy}{4}} =$

(A) 3

(B) 2

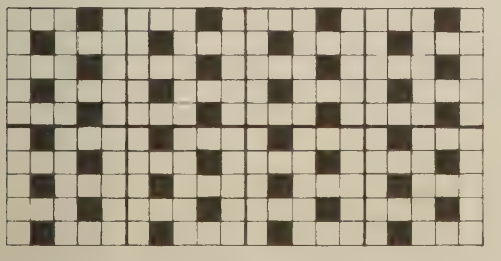
(C) $\frac{4}{5}$

(D) $\frac{1}{2}$

(E) $\frac{17}{36}$

2. Three bananas cost as much as 2 pears. If bananas cost 18 cents each, what is the cost of each pear?

(A) 12¢ (B) 18¢ (C) 24¢
(D) 27¢ (E) 36¢



3. In the figure above, what per cent of the entire figure is shaded?

(A) 5% (B) 20% (C) 25%
(D) 40% (E) $66\frac{2}{3}\%$

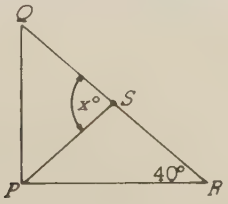


4. If the figure above is the mirror image of an accurate clock, what time will it be 15 minutes after the time shown?

(A) 1:50 (B) 1:40 (C) 1:10
(D) 10:40 (E) 10:05

5. If the sum of 6 consecutive integers is 75, what is the least of these integers?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12



6. If, in $\triangle PQR$ above, $PS = SR$, then $x =$

(A) 50 (B) 80 (C) 90 (D) 100 (E) 110

7. If a rectangular sheet of paper is repeatedly folded in half until the area of the exposed side is $\frac{1}{16}$ of the original area exposed, how many times was the paper folded?

(A) Three (B) Four (C) Five
(D) Six (E) Eight

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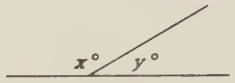
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

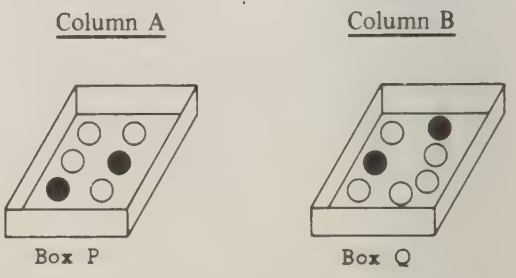
A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. A symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$\frac{1}{2}(24 + 60)$	$\frac{1}{4}(48 + 120)$
9.	$4xy$	$(2x)(2y)$
10.	$2^2 + 3^3$	$3^2 + 2^3$
11.	$x > -3$, $y < 3$: x	y
12.	$x + y = 10$: The average (arithmetic mean) of x and y	5

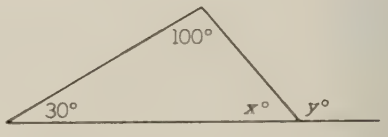


While blindfolded, Jill picked a marble from Box P and a marble from Box Q.

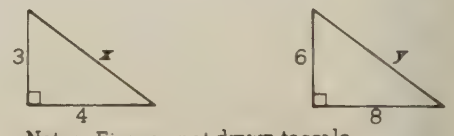
	Column A	Column B
13.	The chance that Jill picked a black marble from Box P	The chance that Jill picked a black marble from Box Q

$X = \{0.98, 0.098, 0.09\}$
$Y = \{0.089, 0.89, 0.9\}$

	Column A	Column B
14.	The greatest number in set X	The greatest number in set Y



	Column A	Column B
15.	$2x$	y



Note: Figures not drawn to scale.

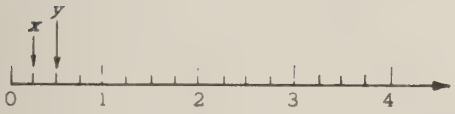
	Column A	Column B
16.	x	$\frac{y}{2}$

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B



x and y are points on the number line.

17. $x \cdot y \cdot 2 \cdot 3 \cdot 4$ | 4

On a certain day, 80 per cent of the girls and 75 per cent of the boys were present in a mathematics class.

18. The number of girls absent | The number of boys absent

---

19. Surface area of a sphere of radius 5 | Area of a circle of radius 5

Questions 20-21 refer to the following definition of $\boxed{x}$ where x is any real number.

$$\boxed{x} = (x-1)^4 + (x-1)^2 + 1$$

20. $\boxed{1}$ | 1

21. $\boxed{10}$ | $\boxed{-10}$

---

22. $\sqrt{5} + \sqrt{11}$ | $\sqrt{16}$

	Column A	Column B

$xy > 0$

23. $x + y$ | $x - y$

Two circles each with radius 4 overlap in such a way that their respective centers P and Q lie in the interior of both circular regions.

24. Length of segment PQ | 4

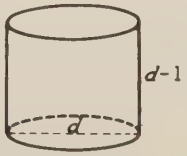
$x^2 - y^2 = 0$

25. x | y

---

x per cent of y is z.
$z \neq 0$

26. 100 | $\frac{xy}{z}$



The diameter of the cylinder is d, where $d > 1$, and the height is $d - 1$.

27. The radius of the largest sphere that will fit completely inside the cylinder | $\frac{d}{2}$

GO ON TO THE NEXT PAGE

5

Solve each of the remaining problems in this section using any available space for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

28. Dividing a number by $\frac{1}{4}$ gives the same result as multiplying by

(A) $\frac{1}{4}$ (B) 0.4 (C) 25% (D) 2 (E) 4

29. Of the following, the closest approximation to $\frac{14{,}995{,}844}{2{,}987}$ is

(A) 500
(B) 700
(C) 5,000
(D) 7,000
(E) 50,000

30. In a certain class of 30 students 18 are girls. If $\frac{2}{3}$ of the girls are either 16 years old or younger, what fractional part of the class is girls over 16 ?

(A) $\frac{2}{15}$

(B) $\frac{1}{5}$

(C) $\frac{1}{3}$

(D) $\frac{2}{5}$

(E) $\frac{1}{2}$

	Scale I	Scale II
Boiling	200°	500°
Freezing	0°	100°

31. The table above shows boiling and freezing points of the same liquid measured under the same conditions on two temperature scales. If the numbers on each of the scales are equally spaced, 50° on scale I would correspond to how many degrees on scale II ?

(A) 125
(B) 150
(C) 200
(D) 225
(E) 250

32. Points P, Q, R, and S lie on a straight line in the order given. The ratio of PQ to QR is 4 to 5 and the ratio of QR to RS is 3 to 7. If $PQ = 12$, what is the length of PS ?

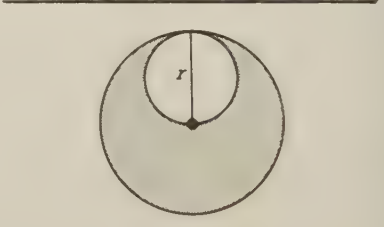
(A) 36
(B) 48
(C) 50
(D) 62
(E) It cannot be determined from the information given.

33. There are 6 baseball teams in a certain league. If in one month each of the 6 teams played exactly 2 games with each of the other teams, how many games were played in the league that month?

(A) 12
(B) 15
(C) 30
(D) 36
(E) 60

34. A metal ball is rolled down a street. Without hitting any bumps, it can travel 200 meters. However, every 20 meters the ball hits a bump and each time the remaining distance it can travel is cut in half. How many bumps will the ball hit before stopping?

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6



35. In the figure above, if one circle has radius r and the other has diameter r, what is the area of the shaded region?

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi r}{4}$ (C) $\frac{\pi r}{2}$

(D) $\frac{\pi r^2}{2}$ (E) $\frac{3\pi r^2}{4}$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 7G027

VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. B	1. D	1. A	1. D
2. A	2. B	2. E	2. D
3. C	3. A	3. B	3. B
4. B	4. A	4. E	4. A
5. A	5. E	5. D	5. C
6. D	6. C	6. E	6. B
7. E	7. B	7. B	7. B
8. E	8. A	8. C	*8. C
9. C	9. C	9. A	*9. C
10. A	10. A	10. A	*10. A
11. A	11. A	11. B	*11. D
12. A	12. D	12. A	*12. C
13. E	13. B	13. E	*13. A
14. B	14. E	14. C	*14. A
15. E	15. E	15. A	*15. B
16. C	16. D	16. B	*16. C
17. D	17. D	17. D	*17. B
18. B	18. A	18. C	*18. D
19. E	19. E	19. B	*19. A
20. D	20. D	20. E	*20. C
21. E	21. A	21. C	*21. B
22. A	22. C	22. E	*22. A
23. C	23. C	23. B	*23. D
24. A	24. E	24. C	*24. B
25. B	25. B	25. D	*25. D
26. B	26. B		*26. C
27. D	27. C		*27. B
28. A	28. E		28. E
29. B	29. D		29. C
30. E	30. A		30. B
31. D	31. C		31. C
32. E	32. E		32. D
33. E	33. C		33. C
34. D	34. A		34. B
35. E	35. C		35. E
36. B	36. B		
37. A	37. A		
38. C	38. D		
39. A	39. C		
40. B	40. C		
41. C			
42. C			
43. E			
44. C			
45. E			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. For example, the SAT-verbal test has 85 questions with five answer choices each, and if a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty between different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing ATP scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the form code you marked in item 3 on your answer sheet is the same as the one printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the opposite page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for section 2 and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover.

FORM CODE 7G027

SAT-Verbal Worksheet

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. Reported SAT-verbal score (See the conversion table on the back cover.) [ ] SAT-Verbal

SAT-Mathematical Worksheet

A. Section 2: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. Reported SAT-mathematical score (See the conversion table on the back cover.) [ ] SAT-Math

If you have any questions on these scoring instructions, you may call the phone number below.

If, after following the above scoring directions and checking your work carefully, your results disagree with the verbal or mathematical score reported on your ATP Student Report, you may request a hand scoring of your answer sheet by writing or calling:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 883-8500

Please indicate whether it is your verbal or mathematical score, or both, that you want to be rescored. When you write, please include a copy of this scoring worksheet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 7G027

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	610
84	790		39	450	600
83	780		38	450	590
82	770		37	440	580
81	760		36	430	580
80	750		35	430	570
79	740		34	420	560
78	730		33	410	550
77	720		32	410	540
76	710		31	400	530
75	710		30	390	520
74	700		29	390	510
73	690		28	380	510
72	680		27	370	500
71	680		26	370	490
70	670		25	360	480
69	660		24	350	470
68	650		23	350	460
67	650		22	340	450
66	640		21	330	440
65	630		20	330	430
64	620		19	320	430
63	620		18	320	420
62	610		17	310	410
61	600		16	300	400
60	590	800	15	300	390
59	590	790	14	290	380
58	580	780	13	280	370
57	570	770	12	280	360
56	570	760	11	270	360
55	560	750	10	260	350
54	550	740	9	260	340
53	550	730	8	250	330
52	540	720	7	240	320
51	530	710	6	240	310
50	530	700	5	230	300
49	520	690	4	220	290
48	510	680	3	220	290
47	510	670	2	210	280
46	500	660	1	200	270
45	490	650	0	200	260
44	490	650	– 1	200	250
43	480	640	– 2	200	240
42	470	630	– 3	200	230
41	470	620	– 4	200	220
			– 5	200	210
			– 6	200	210
			– 7 or below	200	200